PostgreSQL – Architektur

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Zusammenstellung von Folien aus einem EDB-Course

PostgreSQL

The open-source database of choice

Performance

Handles enterprise workloads with 50% improvement in the last 4 years

Scalability

Multiple technical options for operating Postgres at scale

Extensibility

Supported by a wide array of extensions plus multiple SQL and NoSQL data models

Community-driven

Multiple companies and individuals contribute to the project and drive innovation

Facts about PostgreSQL

The world's most advanced open-source database

Designed for extensibility and customization

ANSI/ISO compliant SQL support

Actively developed for more almost 40 years

- University Postgres (1986-1993)
- Postgres95 (1994-1995)
- PostgreSQL (1996-current)

Major Features

Portable:

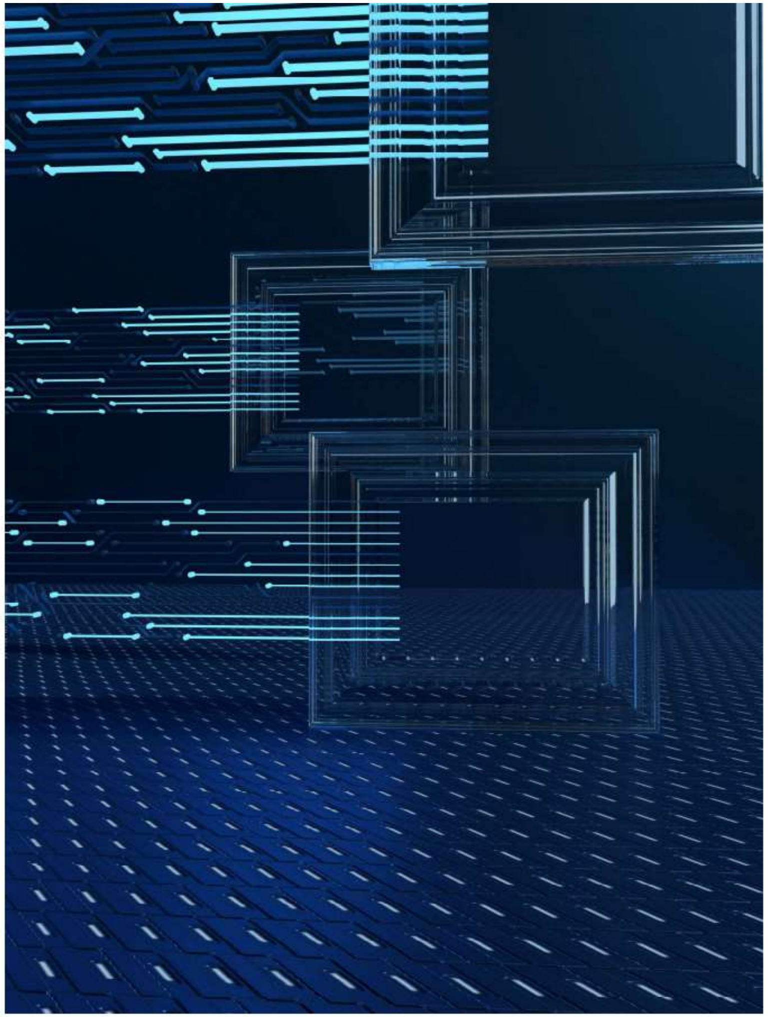
- Written in ANSI C
- Supports Windows, Linux, Mac OS/X and major UNIX platforms

Reliable:

- ACID Compliant
- Supports Transactions and Savepoints
- Uses Write Ahead Logging (WAL)

Scalable:

- Uses Multi-version Concurrency Control
- Table Partitioning and Tablespaces
- Parallel Sequential Scans, DDL(Table and Index Creation)



Major Features (continued)

Secure:

- Employs Host-Based Access Control
- Provides Object-Level Permissions and Row Level Security
- Supports SSL Connections and Logging
- Transparent Data Encryption - TDE

Recovery and Availability:

- Streaming Replication, Logical Replication and Replication Slots
- Replication Slots, Sync or Async Options
- Supports Hot-Backup, pg_basebackup, incremental backup and Point-in-Time Recovery

Advanced:

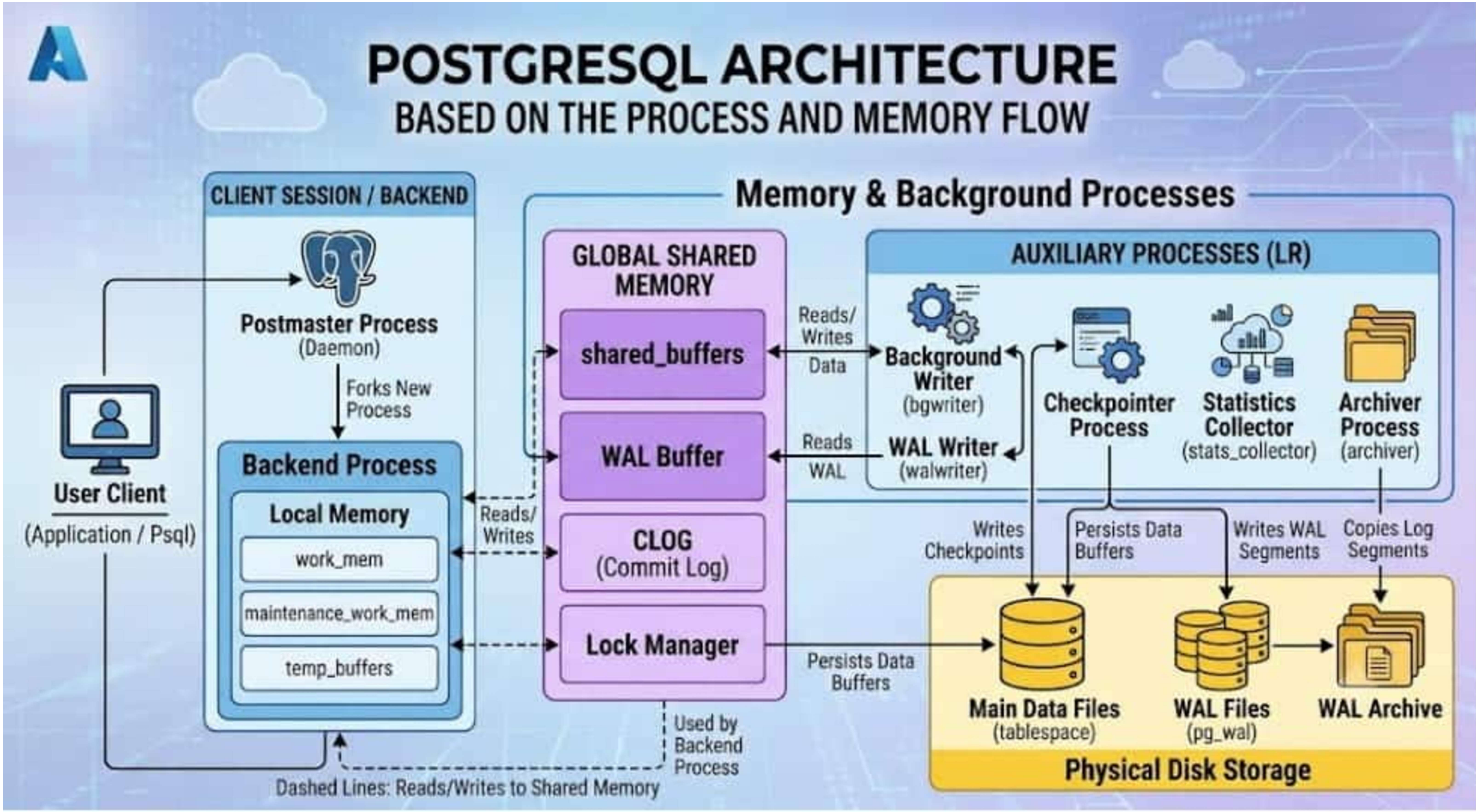
- Supports Triggers, Functions and Procedures
- Supports Custom Procedural Languages
- Upgrade using pg_upgrade
- Unlogged Tables and Materialized Views
- Just-in-Time (JIT) Compilation

General Database Limits

Limit	Value
Maximum Database Size	Unlimited
Maximum Table Size	32 TB
Maximum Row Size	1.6 TB
Maximum Field Size	1 GB
Maximum Rows per Table	Unlimited
Maximum Columns per Table	250-1600 (Depending on Column types)
Maximum Indexes per Table	Unlimited

Common Database Object Names

Industry Term	Postgres Term
Table or Index	Relation
Row	Tuple
Column	Attribute
Data Block	Page (when block is on disk)
Page	Buffer (when block is in memory)



POSTGRESQL ARCHITECTURE
BASED ON THE PROCESS AND MEMORY FLOW
CLIENT SESSION / BACKEND
Postmaster Process
(Daemon)
Forks New Process
Backend Process
Local Memory
work_mem
maintenance_work_mem
temp_buffers
User Client
(Application / Psql)
Memory & Background Processes
GLOBAL SHARED MEMORY
shared_buffers
WAL Buffer
CLOG
(Commit Log)
Lock Manager
Reads/ Writes
Reads/ Writes Data
Reads WAL
AUXILIARY PROCESSES (LR)
Background Writer
(bgwriter)
WAL Writer
(walwriter)
Checkpointer Process
Statistics Collector
(stats_collector)
Archiver Process
(archiver)
Writes Checkpoints
Persists Data Buffers
Writes WAL Segments
Copies Log Segments
Persists Data Buffers
Main Data Files
(tablespace)
WAL Files
(pg_wal)
WAL Archive
Physical Disk Storage
Used by Backend Process
Dashed Lines: Reads/Writes to Shared Memory

Architectural Summary

Postgres uses **processes**, not threads

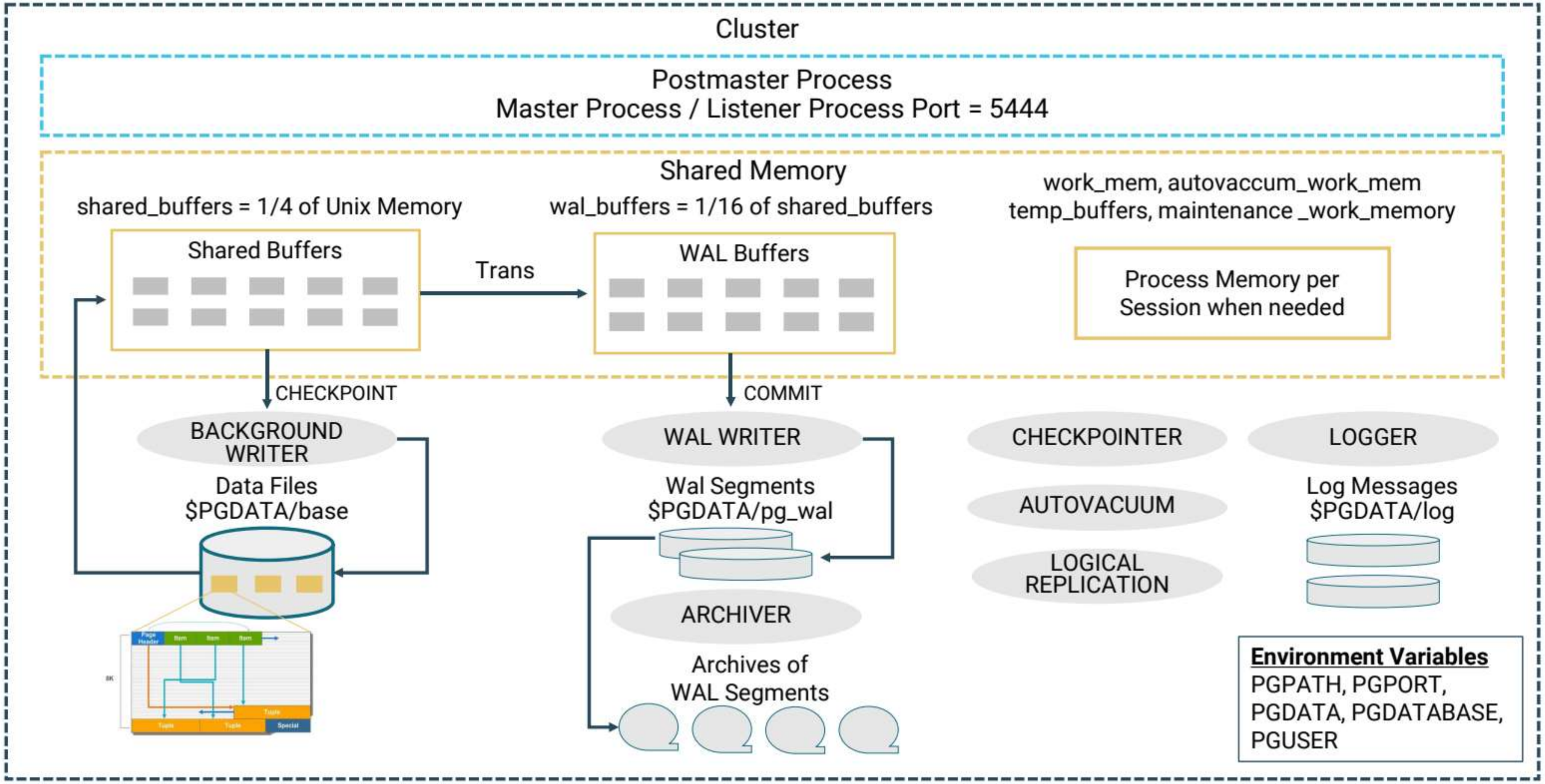
The “Postmaster” process acts as a **supervisor**

Several utility processes perform **background** work

- postmaster starts them and restarts them if they die

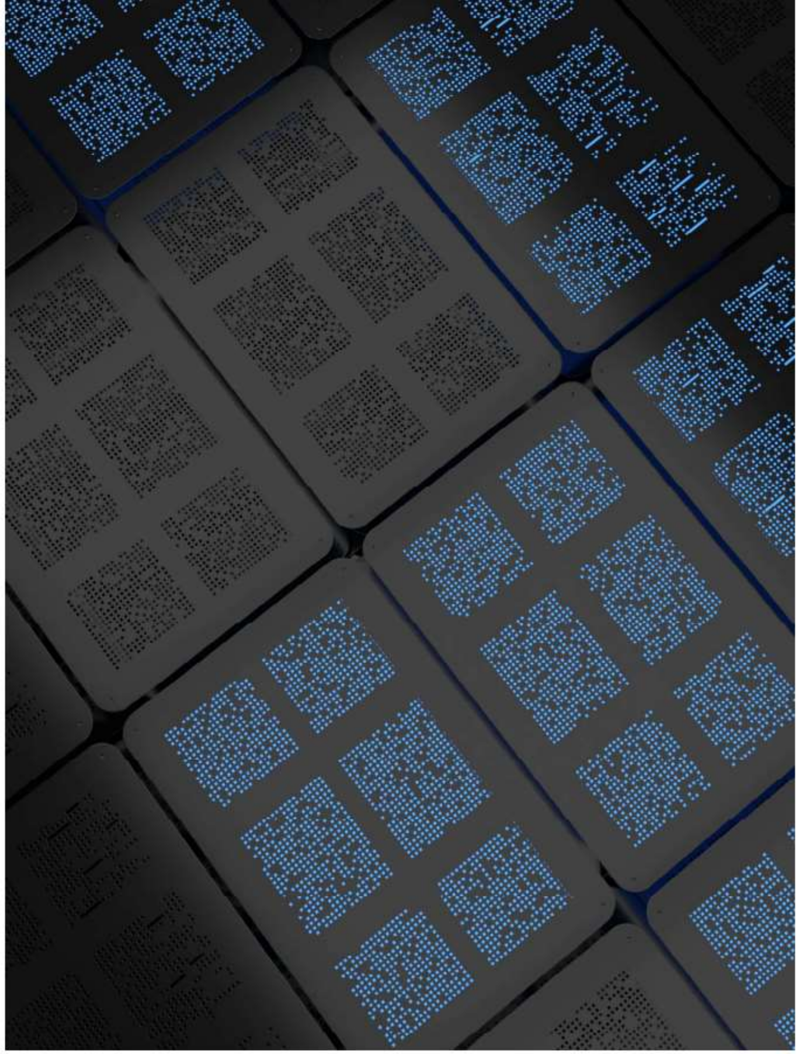
One backend process is created **per user session**

- postmaster listens for new connections


Cluster
Postmaster Process
Master Process / Listener Process Port = 5444
Shared Memory
shared_buffers = 1/4 of Unix Memory
wal_buffers = 1/16 of shared_buffers
work_mem, autovaccum_work_mem
temp_buffers, maintenance _work_memory
Shared Buffers
Trans
WAL Buffers
Process Memory per
Session when needed
CHECKPOINT
COMMIT
BACKGROUND
WRITER
WAL WRITER
CHECKPOINTER
LOGGER
Data Files
$PGDATA/base
Wal Segments
$PGDATA/pg_wal
AUTOVACUUM
Log Messages
$PGDATA/log
LOGICAL
REPLICATION
ARCHIVER
Archives of
WAL Segments
Environment Variables
PGPATH, PGPORT,
PGDATA, PGDATABASE,
PGUSER

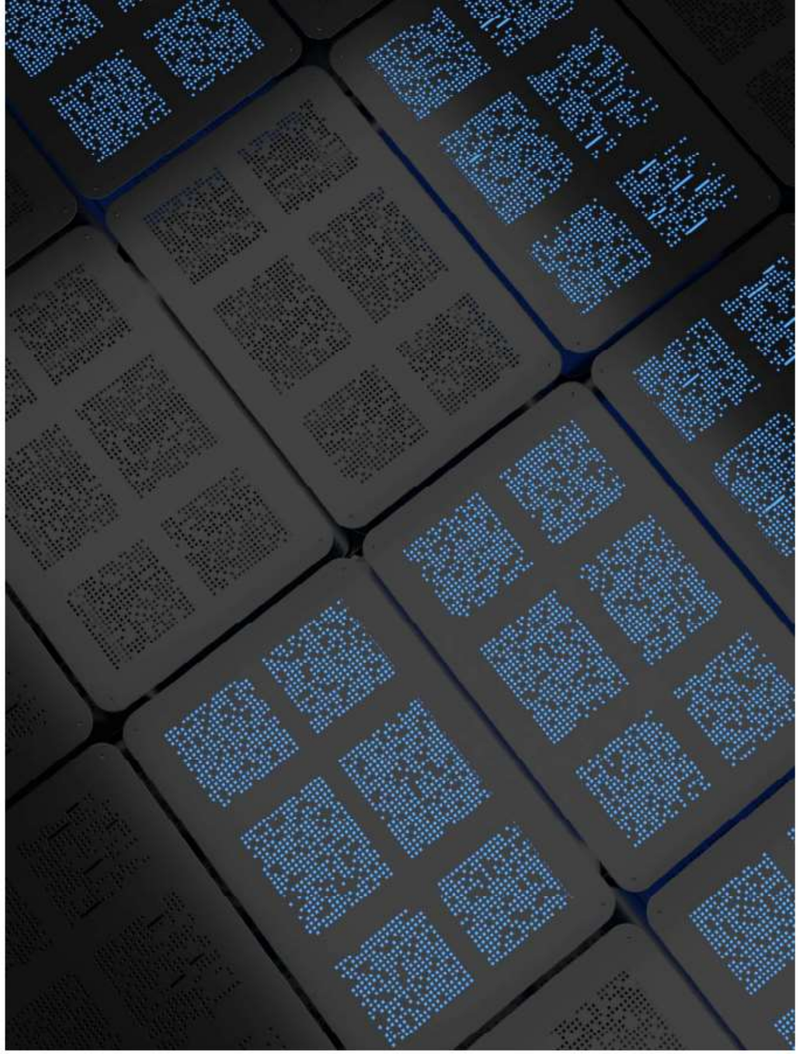
Utility Processes

- Background writer
  - Writes dirty data blocks to disk
- WAL writer
  - Flushes write-ahead log to disk
- Checkpointer
  - Automatically performs a checkpoint based on config parameters
- Autovacuum launcher
  - Starts Autovacuum workers as needed
- Autovacuum workers
  - Recover free space for reuse



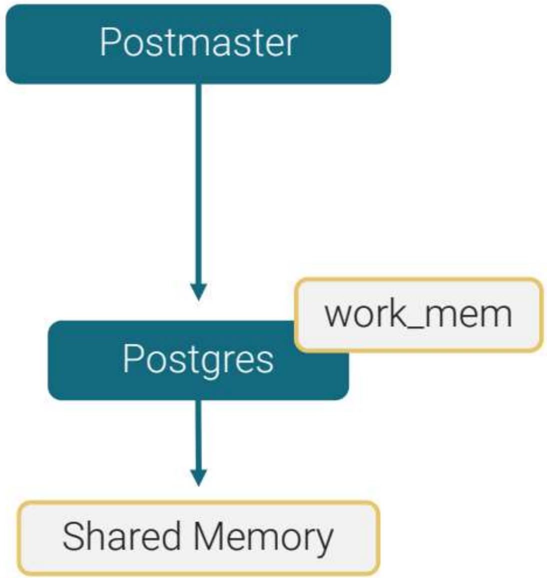
More Utility Process

- Logger - Logging collector
  - Routes log messages to syslog, eventlog, or log files
- Archiver
  - Archives write-ahead log files
- Logical replication launcher
  - Starts logical replication apply process for logical replication
- Dbms_aq launcher
  - Collects information for queueing functionality of advanced server



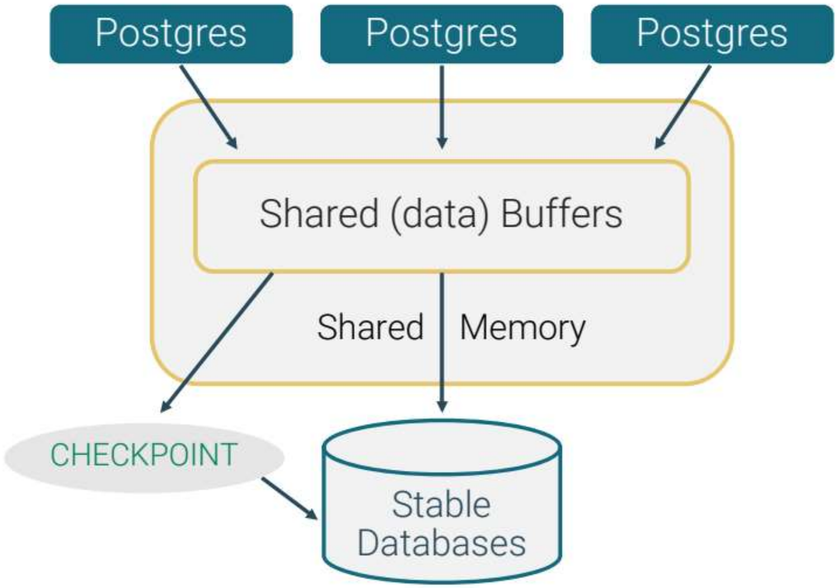
User Backend Process

- Postmaster process spawns a new server process for each connection request detected
- Communication is done using semaphores and shared memory
- Authentication - IP, user and password
- Authorization - Verify permissions



Disk Write Buffering

- Blocks are written to disk only when needed:
  - To make room for new blocks
  - At checkpoint time



Commit and Checkpoint

Before commit	Uncommitted updates are in memory
After commit	WAL buffers are written to the disk (write-ahead log file) and shared buffers are marked as committed
After checkpoint	Modified data pages are written from shared memory to the data files

Statement Processing

Parse

- Check syntax
- Call traffic cop
- Identify query type
- Command processor if needed
- Break query into tokens

Optimize

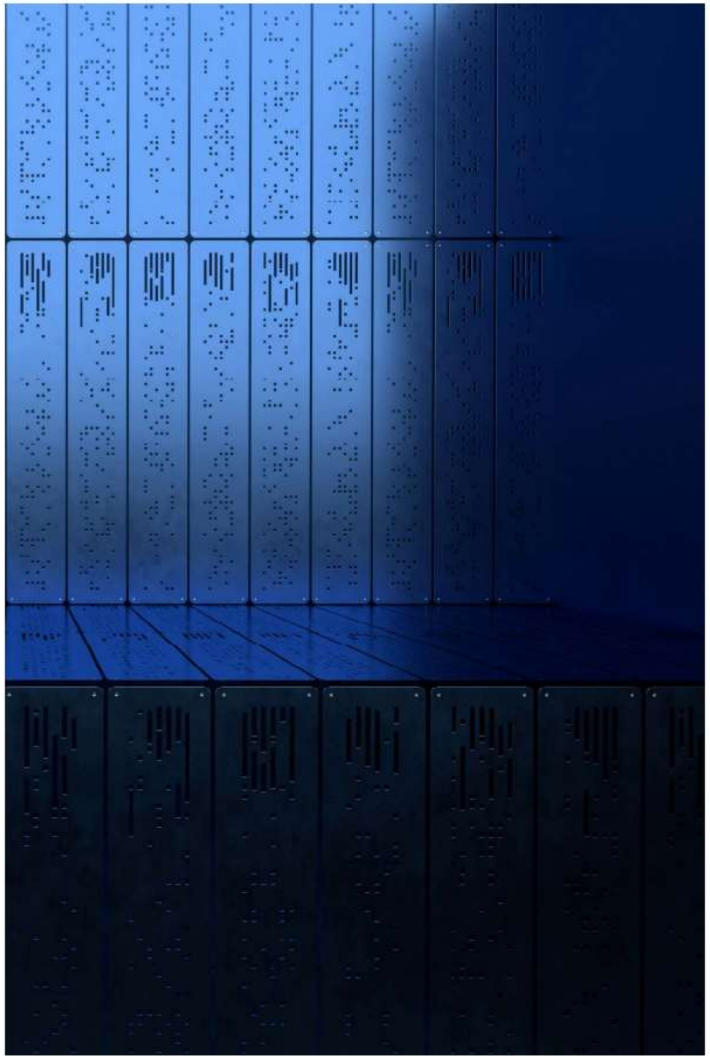
- Planner generates a plan
- Uses database statistics
- Apply Optimizer Hints
- Query cost calculation
- Choose best plan

Execute

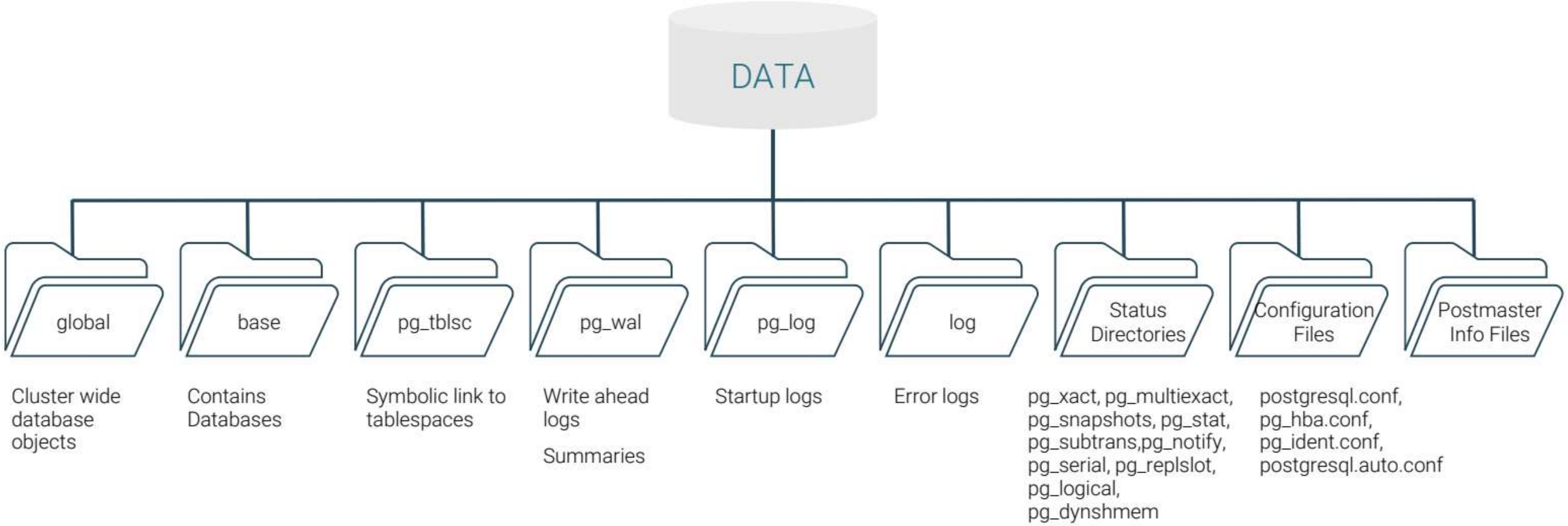
- Execute query based on query plan

Physical Database Architecture

- Database cluster is a collection of databases managed by single server instance
- Each cluster has a separate
  - Data directory
  - TCP port
  - Set of processes
- A cluster can contain multiple databases



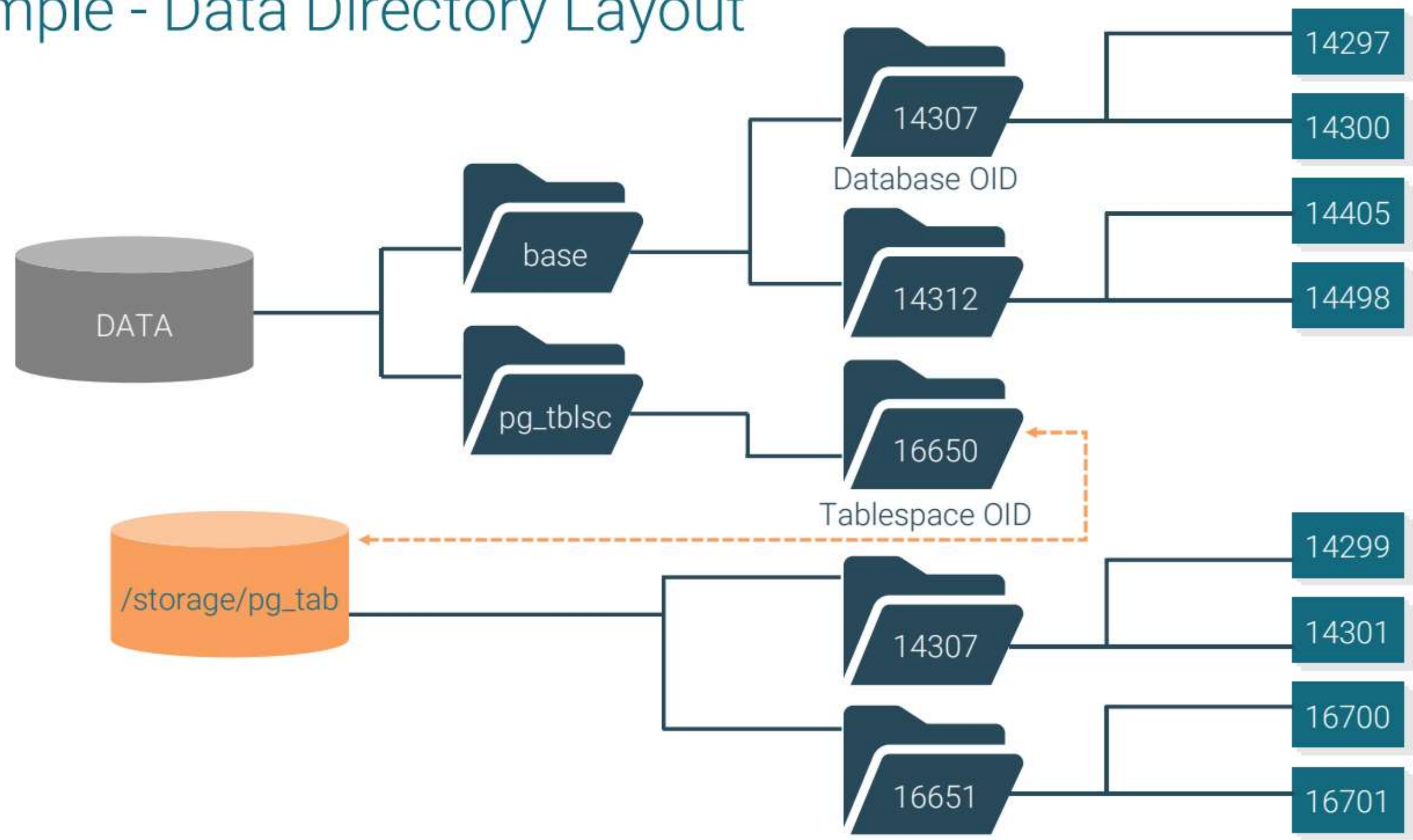
Database Cluster Data Directory Layout



Physical Database Architecture

- File-per-table, file-per-index
- A table-space is a directory
- Each database that uses that table-space gets a subdirectory
- Each relation using that table-space/database combination gets one or more files, in 1GB chunks
- Additional files are used to hold auxiliary information (free space map, visibility map)
- Each file name is a number (see **pg_class.relfilenode**)

Sample - Data Directory Layout



Introduction to psql

- psql is a command line interface (CLI) to Postgres
- Can be used to execute SQL queries and psql meta commands

```
[enterprisedb@Base ~]$ psql -h /tmp -p 5444 -U enterprisedb -d edb
Type "help" for help.
edb=# \q
```

Connecting to a Database

psql Connection Options:

- -d <Database Name>
- -h <Hostname>
- -p <Database Port>
- -U <Database Username>

Environmental Variables

- PGDATABASE, PGHOST, PGPORT and PGUSER